

# Malhar Gudekar

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## EDUCATION

### University of Illinois Urbana Champaign

August 2024 - May 2026

Master of Science, Information Management

GPA: 3.8/4

• Coursework: Web Programming, Machine Learning & Cloud, Information Management, Information Consulting, Methods of Data Science

### University of Mumbai

June 2019 - May 2023

Bachelor of Engineering, Computer Science

GPA: 9.25/10

• Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, Object-Oriented Programming, Machine Learning, NLP

## SKILLS

• **Languages:** Python, SQL, Java(leetcode), C++(leetcode)

• **Data Engineering:** ETL/ELT Pipelines, Data Modeling, Snowflake, PySpark, Workflow Orchestration (Airflow, Step Functions, AWS Glue)

• **Cloud & Tools:** AWS (S3, EC2, EKS), PostgreSQL, Neo4j, Git, CI/CD, Docker

## WORK EXPERIENCE

### University of Illinois Urbana Champaign

May 2025 - Present

*Research Assistant - iSchool*

Champaign, USA

• Built and maintained AWS data pipelines across EC2, S3, and VPC-secured network environments to ingest high-volume data for 100+ users, enabling scalable distributed storage and downstream analytics

• Designed and optimized **SQL-based data models and warehouse tables** in PostgreSQL, applying indexing and query tuning techniques to improve performance by 38% for large datasets

• Implemented workflow orchestration using AWS Step Functions with CloudWatch monitoring, configuring IAM roles with least-privilege access and network security controls to ensure reliable and secure pipeline operation

*Research Assistant - CHI AI in Infodemic Management*

January 2025 - May 2025

• Designed and deployed **scalable ELT pipelines** in **Python** for high-volume social media datasets in distributed environments, reducing processing latency by 41%

• Built distributed real-time ingestion pipelines using **Kafka** and **Neo4j**, enabling streaming data processing, transformation, and graph-based analytics at scale

• Applied **distributed data processing and big data principles** to handle large-scale streaming data and ensure consistent data availability

### Business Intelligence Group

August 2025 - December 2025

*Technical Consultant*

Champaign, USA

• Built and deployed scalable **data processing pipelines** using **FastAPI**, **PostgreSQL**, and optimized storage layers to support large-scale analytics and downstream applications

• Designed fault-tolerant ETL workflows with documented runbooks for ingesting and transforming high-volume datasets, ensuring low-latency data availability and clear operational handoffs

• Developed backend services using Git-based CI/CD workflows, monitoring system health with CloudWatch dashboards and responding to incidents to maintain uptime and meet SLA requirements

### Swift Mobil Software Solutions Provider

July 2023 - October 2023

*Data Analyst*

Mumbai, IN

• Built **data pipelines** feeding **Power BI dashboards** for logistics KPIs, reducing reporting time by **40%** and improving data availability for business decision-making

• Wrote Python and Bash scripts to automate data processing, log parsing, and system health checks, reducing manual operational toil and improving performance monitoring visibility

• Developed **SQL queries** to validate, reconcile, and analyze operational data, improving data quality and speeding root-cause analysis by **26%**

### Pscope Technologies Pvt. Ltd.

January 2023 - June 2023

*Data Analyst*

Mumbai, IN

• Built and maintained **SQL-based data pipelines** supporting **Power BI dashboards**, designing data warehouse models, writing complex joins and aggregations, and ensuring delivery of analytics-ready datasets across business functions

• Automated **data transformation workflows** using VBA, implementing validation checks, error handling, and scheduling logic to streamline large-scale data processing and cut reporting time by **42%**

### Stu/Dio at Illinois

August 2025 - Present

*Project Manager*

Champaign, USA

• Led cross-functional team of **6** engineers to deliver data-driven improvements, analytics insights, and system tracking

• Managed Agile workflows using JIRA, documenting issues, resolutions, and runbooks clearly to support incident tracking, sprint progress, and cross-team handoffs during delivery cycles

## PROJECTS

### Scalable ML System for Customer Risk Prediction

March 2025

UIUC Hackathon

• Built an **end-to-end data pipeline** using Python, Pandas, and LightGBM to process large-scale **time-series** transaction data, including ingestion, transformation, feature engineering, and model integration

• Designed and implemented **scalable ETL and feature engineering pipelines** for large datasets, capturing behavioral trends and temporal patterns to improve model performance

• Developed a **data-driven decision framework**, incorporating pipeline outputs and risk thresholds to improve prediction precision by 20% and enable scalable analytics workflows